

H. Differential Inductive Moment Matching

Similar to the continuous-time CMs presented in (Lu & Song, 2024), our MMD objective can be taken to the differential limit. Consider the simplified loss and parameterization in Eq. (12), we use the RBF kernel as our kernel of choice for simplicity.

Theorem 2 (Differential Inductive Moment Matching). *Let $\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$ be a twice continuously differentiable function with bounded first and second derivatives, let $k(\cdot, \cdot)$ be RBF kernel with unit bandwidth, $\mathbf{x}, \mathbf{x}' \sim q(\mathbf{x})$, $\mathbf{x}_t \sim q_t(\mathbf{x}_t|\mathbf{x})$, $\mathbf{x}'_t \sim q_s(\mathbf{x}'_t|\mathbf{x}')$, $\mathbf{x}_r = \text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, r)$ and $\mathbf{x}'_r = \text{DDIM}(\mathbf{x}'_t, \mathbf{x}', t, r)$, the following objective*

$$\mathcal{L}_{\text{IMM-}\infty}(\theta, t) = \lim_{r \rightarrow t} \frac{1}{(t-r)^2} \mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t, \mathbf{x}_r, \mathbf{x}'_r} \left[k\left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)\right) + k\left(\mathbf{f}_{s,r}^\theta(\mathbf{x}_r), \mathbf{f}_{s,r}^\theta(\mathbf{x}'_r)\right) - k\left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,r}^\theta(\mathbf{x}'_r)\right) - k\left(\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t), \mathbf{f}_{s,r}^\theta(\mathbf{x}_r)\right) \right] \quad (104)$$

can be analytically derived as

$$\mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t} \left[e^{-\frac{1}{2} \|\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)\|^2} \left(\frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt}^\top \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)}{dt} - \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt}^\top \left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right) \left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right)^\top \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)}{dt} \right) \right] \quad (105)$$

Proof. Firstly, the limit can be exchanged with the expectation due to dominated convergence theorem where the integrand consists of kernel functions which can be assumed to be upper bounded by 1, e.g. RBF kernels are upper bounded by 1, and thus integrable. It then suffices to check the limit of the integrand. Before that, let us review the first and second-order Taylor expansion of $e^{-\|\mathbf{x}-\mathbf{y}\|^2}$. We let $a, b \in \mathbb{R}^D$ be constants to be expanded around. We note down the Taylor expansion to second-order below for notational convenience.

- $e^{-\|\mathbf{x}-\mathbf{b}\|^2/2}$ at $\mathbf{x} = \mathbf{a}$:

$$e^{-\|\mathbf{a}-\mathbf{b}\|^2/2} - e^{-\|\mathbf{a}-\mathbf{b}\|^2/2} (\mathbf{a}-\mathbf{b})^\top (\mathbf{x}-\mathbf{a}) + \frac{1}{2} (\mathbf{x}-\mathbf{a})^\top e^{-\|\mathbf{a}-\mathbf{b}\|^2/2} ((\mathbf{a}-\mathbf{b})(\mathbf{a}-\mathbf{b})^\top - I) (\mathbf{x}-\mathbf{a})$$

- $e^{-\|\mathbf{y}-\mathbf{a}\|^2/2}$ at $\mathbf{y} = \mathbf{b}$:

$$e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} - e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} (\mathbf{b}-\mathbf{a})^\top (\mathbf{y}-\mathbf{b}) + \frac{1}{2} (\mathbf{y}-\mathbf{b})^\top e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} ((\mathbf{b}-\mathbf{a})(\mathbf{b}-\mathbf{a})^\top - I) (\mathbf{y}-\mathbf{b})$$

- $e^{-\|\mathbf{x}-\mathbf{y}\|^2/2}$ at $\mathbf{x} = \mathbf{a}, \mathbf{y} = \mathbf{b}$:

$$\begin{aligned} & e^{-\|\mathbf{a}-\mathbf{b}\|^2/2} - e^{-\|\mathbf{a}-\mathbf{b}\|^2/2} (\mathbf{a}-\mathbf{b})^\top (\mathbf{x}-\mathbf{a}) - e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} (\mathbf{b}-\mathbf{a})^\top (\mathbf{y}-\mathbf{b}) \\ & + \frac{1}{2} (\mathbf{y}-\mathbf{b})^\top e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} ((\mathbf{b}-\mathbf{a})(\mathbf{b}-\mathbf{a})^\top - I) (\mathbf{y}-\mathbf{b}) \\ & + \frac{1}{2} (\mathbf{y}-\mathbf{b})^\top e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} ((\mathbf{b}-\mathbf{a})(\mathbf{b}-\mathbf{a})^\top - I) (\mathbf{y}-\mathbf{b}) \\ & + (\mathbf{x}-\mathbf{a})^\top e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} (I - (\mathbf{a}-\mathbf{b})(\mathbf{a}-\mathbf{b})^\top) (\mathbf{y}-\mathbf{b}) \end{aligned}$$

Putting it together, the above results imply

$$\begin{aligned} & e^{-\|\mathbf{x}-\mathbf{y}\|^2/2} + e^{-\|\mathbf{a}-\mathbf{b}\|^2/2} - e^{-\|\mathbf{x}-\mathbf{b}\|^2/2} - e^{-\|\mathbf{y}-\mathbf{a}\|^2/2} \\ & \approx (\mathbf{x}-\mathbf{a})^\top e^{-\|\mathbf{b}-\mathbf{a}\|^2/2} (I - (\mathbf{a}-\mathbf{b})(\mathbf{a}-\mathbf{b})^\top) (\mathbf{y}-\mathbf{b}) \end{aligned}$$

since it is easy to check that the remaining terms cancel.

Substituting $x = \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$, $a = \mathbf{f}_{s,r}^\theta(\mathbf{x}_r)$, $y = \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)$, $b = \mathbf{f}_{s,r}^\theta(\mathbf{x}'_r)$, we furthermore have

$$\lim_{r \rightarrow t} \frac{1}{t-r} (x-a) = \lim_{r \rightarrow t} \frac{1}{t-r} [\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,r}^\theta(\mathbf{x}_r)] \quad (106)$$

$$= \lim_{r \rightarrow t} \frac{1}{t-r} \left[(t-r) \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt} + \mathcal{O}(|t-r|^2) \right] \quad (107)$$

$$= \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt} \quad (108)$$

Similarly,

$$\lim_{r \rightarrow t} \frac{1}{t-r} (y-b) = \lim_{r \rightarrow t} \frac{1}{t-r} [\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) - \mathbf{f}_{s,r}^\theta(\mathbf{x}'_r)] = \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)}{dt} \quad (109)$$

Therefore, $\mathcal{L}_{\text{IMM-}\infty}(\theta, t)$ can be derived as

$$\mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t} \left[e^{-\frac{1}{2} \|\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)\|^2} \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt}^\top \right] \quad (110)$$

$$\left(I - \left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right) \left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right)^\top \right) \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)}{dt} \right] \quad (111)$$

$$= \mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t} \left[e^{-\frac{1}{2} \|\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)\|^2} \left(\frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt}^\top \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)}{dt} \right. \right. \quad (112)$$

$$\left. \left. - \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt}^\top \left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right) \left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right)^\top \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)}{dt} \right) \right]$$

□

H.1. Pseudo-Objective

Due to the stop-gradient operation, we can similarly find a *pseudo-objective* whose gradient matches the gradient of $\mathcal{L}_{\text{IMM-}\infty}(\theta, t)$ in the limit of $r \rightarrow t$.

Theorem 3. Let $\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$ be a twice continuously differentiable function with bounded first and second derivatives, $k(\cdot, \cdot)$ be RBF kernel with unit bandwidth, $\mathbf{x}, \mathbf{x}' \sim q(\mathbf{x})$, $\mathbf{x}_t \sim q_t(\mathbf{x}_t|\mathbf{x})$, $\mathbf{x}'_t \sim q_t(\mathbf{x}'_t|\mathbf{x}')$, $\mathbf{x}_r = \text{DDIM}(\mathbf{x}_t, \mathbf{x}, t, r)$ and $\mathbf{x}'_r = \text{DDIM}(\mathbf{x}'_t, \mathbf{x}', t, r)$, the gradient of the following pseudo-objective

$$\nabla_\theta \mathcal{L}_{\text{IMM-}\infty}^-(\theta, t) = \lim_{r \rightarrow t} \frac{1}{(t-r)} \nabla_\theta \mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t, \mathbf{x}_r, \mathbf{x}'_r} \left[k\left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)\right) + k\left(\mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r), \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r)\right) \right. \quad (113)$$

$$\left. - k\left(\mathbf{f}_{s,t}^\theta(\mathbf{x}_t), \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r)\right) - k\left(\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t), \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r)\right) \right]$$

can be used to optimize θ and can be analytically derived as

$$\mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t} \left[e^{-\frac{1}{2} \|\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)\|^2} \left(\left(\frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)}{dt}^\top - \left[\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t) \right]^\top \frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)}{dt} \left[\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t) \right]^\top \right) \nabla_\theta \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) + \right. \quad (114)$$

$$\left. \left(\frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt}^\top - \left[\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t) \right]^\top \frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt} \left[\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t) \right]^\top \right) \nabla_\theta \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right]$$

Proof. Similar to the derivation of $\mathcal{L}_{\text{IMM-}\infty}(\theta, t)$, let $x = \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$, $a = \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}_r)$, $y = \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t)$, $b = \mathbf{f}_{s,r}^{\theta^-}(\mathbf{x}'_r)$, we have

$$\begin{aligned} & \lim_{r \rightarrow t} \frac{1}{(t-r)} \nabla_\theta (x-a)^\top e^{-\|b-a\|^2/2} (I - (a-b)(a-b)^\top) (y-b) \\ &= \lim_{r \rightarrow t} \frac{1}{(t-r)} e^{-\|b-a\|^2/2} \left[\left((y-b)^\top - ((a-b)^\top (y-b)) (a-b)^\top \right) \nabla_\theta x + \right. \\ & \quad \left. \left((x-a)^\top - ((a-b)^\top (x-a)) (a-b)^\top \right) \nabla_\theta y \right] \end{aligned}$$

where

$$\lim_{r \rightarrow t} \frac{1}{(t-r)} (x-a) = \frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt}, \quad \lim_{r \rightarrow t} \frac{1}{(t-r)} (y-b) = \frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)}{dt}$$

Note that $\frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt}$ is now parameterized by θ^- instead of θ because the gradient is already taken w.r.t. $\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$ outside of the brackets, so $(x-a)$ and $(y-b)$ merely require *evaluation* at current θ with no gradient information, which θ^- satisfies. The objective can be derived as

$$\begin{aligned} & \nabla_\theta \mathcal{L}_{\text{IMM-}\infty}^-(\theta, t) \\ &= \mathbb{E}_{\mathbf{x}_t, \mathbf{x}'_t} \left[e^{-\frac{1}{2} \|\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)\|^2} \left(\right. \right. \\ & \quad \left(\frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)}{dt} - [\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)]^\top \frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)}{dt} [\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)]^\top \right) \nabla_\theta \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) + \\ & \quad \left. \left(\frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt} - [\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)]^\top \frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt} [\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t) - \mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}'_t)]^\top \right) \nabla_\theta \mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \right) \right] \end{aligned}$$

□

H.2. Connection with Continuous-Time CMs

Observing Eq. (105) and Eq. (110), we can see that when $\mathbf{x}_t = \mathbf{x}'_t$ and $\mathbf{x}_r = \mathbf{x}'_r$, s being a small positive constant, then $\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) = \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)$, and $\exp\left(-\frac{1}{2} \|\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) - \mathbf{f}_{s,t}^\theta(\mathbf{x}_t)\|^2\right) = 1$, and $\mathbf{f}_{s,t}^\theta(\mathbf{x}'_t) \approx \mathbf{g}_\theta(\mathbf{x}_t, t)$ where since s is fixed we discard the dependency on s as input. Then, Eq. (105) reduces to

$$\mathbb{E}_{\mathbf{x}_t} \left[\left\| \frac{d\mathbf{f}_{s,t}^\theta(\mathbf{x}_t)}{dt} \right\|^2 \right] \quad (115)$$

which is the same as differential consistency loss (Song et al., 2023; Geng et al., 2024). And Eq. (110) reduces to

$$\mathbb{E}_{\mathbf{x}_t} \left[\frac{d\mathbf{f}_{s,t}^{\theta^-}(\mathbf{x}_t)}{dt}^\top \nabla_\theta \mathbf{f}_{s,t}^\theta(\mathbf{x}_t) \right] \quad (116)$$

which is the pseudo-objective for continuous-time CMs (Song et al., 2023; Lu & Song, 2024) (minus a weighting function of choice).

I. Experiment Settings

I.1. Training & Parameterization Settings

We summarize our best runs in Table 5. Specifically, for ImageNet-256×256, we adopt a latent space paradigm for computational efficiency. For its autoencoder, we follow EDM2 (Karras et al., 2024) and pre-encode all images from